D.2 SFT Details

Algorithm 2 Supervised Finetuning of LLaDA [30]

Require: underlying unmasking predictor f_θ, data distribution p_{data}, learning rate η

repeat
  Sample $(p_0, r_0) \sim p_{\text{data}}, \quad t \sim \mathcal{U}(0,1)$ ▷ p_0 is the prompt and r_0 is the response
  Construct a partially masked response $r_t \sim q_{t|0}(r_t|r_0)$ ▷ $q_{t|0}$ is defined in Eq. (5)
  Calculate $\mathcal{L}(\theta) = -\frac{1}{t|r_0|}\sum_{i=1}^{|r_0|} \mathbb{1}[r_t^i = \texttt{mask}] \log f_\theta(r_0^i|p_0 \oplus r_t)$ ▷ $\oplus$ is concatenation
  $\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}$
until Converged
Return θ

Similarly, the SFT model also employs LoRA, with a rank of $r = 128$ and scaling factor $\alpha = 256$. We train with a sequence length of 4096 on 2 A6000 GPUs, using gradient accumulation over 4 steps and a per-device batch size of 1, yielding an effective batch size of 8. The optimizer and learning rate schedule match those used in *diffu*-GRPO, with a learning rate of 1e-5 and gradient clipping at 1.0. The SFT model was trained on the s1k dataset for 2460 steps, leaving 1% of the data for evaluation. A linear learning rate decay schedule was used, with no warmup. Our codebase contains further configuration details: `https://github.com/dllm-reasoning/d1`.

Truncated Sequences LLaDA-instruct is trained to generate full sentences, i.e., given any sequence length, it will always try to generate a complete sentence. However, due to the long sequence length of s1k, we had to truncate the dataset to have a maximum sequence length of 4096.

Loss on PAD tokens As discussed in Nie et al. [30], LLaDA needs to take a loss on the PAD tokens to be able to effectively terminate its generation. Additionally, to speed up training, we can pad the sequences in a batch to the longest sequence length in the batch. However, in GPU-constrained environments which use a small batch size, we find that padding to the longest datapoint in the batch is suboptimal since not enough PAD tokens are considered in the loss. Therefore, we choose to pad to the max length of the model.

Dataset Difficulty We find that there are some dynamics between the difficulty of the dataset and the model strength. In general, a weaker model needs a combination of easier and harder datapoints to scale reasoning. If we use an overly hard dataset, the performance degrades since the model easily overfits.

E Qualitative Examples on GSM8k

We show qualitative results from all the models. Correct answers are in green, incorrect answers in red and aha-moments in blue. Notably, sometimes the model can reach the right answer via the wrong reasoning - such artifacts should reduce with a stronger base model.

E.1 LLaDA-Instruct

When controlling the generation sequence length of dLLMs, it influences the reasoning style. For example, we see that with seq len = 512, compared to 128, the model produces more detailed step-by-step solutions with explicit mathematical formulas and enumerated calculations.

Model: LLaDA-8B-Instruct
Generation Sequence Length: 128
Question: A 76-star flag has three rows of 8 stars two rows of 6 stars and the rest are 5-star rows. How many rows of 5 stars are there altogether on the flag?

<reasoning> The flag has three stars in 8 rows, two stars in 6 rows, and the rest are 5-star rows. The total number of stars is the sum of the stars in the three8-star rows and the 6-star rows, which is 24 + 12 = 36 stars. Therefore, the remaining 76 - 36 = 40 stars are be placed in 5-star rows. Hence, the number of 5-star rows is 40/5. < /reasoning> <answer> $\boxed{8}$ < /answer>

Model: LLaDA-8B-Instruct
Generation Sequence Length: 512
Question: A 76-star flag has three rows of 8 stars two rows of 6 stars and the rest are 5-star rows. How many rows of 5 stars are there altogether on the flag?

<reasoning> To determine the number of rows of 5 stars on the 76-star flag, we need to follow these steps:
1. Calculate the total number of stars in the three rows of 8 stars. 2. Calculate the total number of stars in the two rows of 6 stars. 3. Subtract the total number of stars in the first two rows from the total number of stars on the flag to find the number of stars in the 5-star rows. 4. Divide the number of stars in the 5-star rows by 5 to find the number of 5-star rows rows.
Let's go through each step in detail:
1. The total number of stars in the three rows of 8 stars is:

$$3 \times 8 = 24$$

2. The total number of stars in the two rows of 6 stars is:

$$2 \times 6 = 12$$

3. The total number of stars in the first two rows is:

$$24 + 12 = 36$$

4. The total number of stars on the flag is 76. Therefore, the number of stars in the 5-star rows is:

$$76 - 36 = 40$$

5. The number of 5-star rows is the number of stars in the 5-star rows divided by 5:

$$\frac{40}{5} = 8$$

So, the number of rows of 5 stars on the flag is $\boxed{8}$ < /reasoning>

Model: LLaDA-8B-Instruct
Generation Sequence Length: 512
Question: A bakery produces 60 loaves of bread each day. Two-thirds of the loaves are sold in the morning and half of what is left is sold equally in the afternoon and evening. How many loaves of bread are sold in the afternoon?

<reasoning> To determine the number of loaves of bread sold in the afternoon, we need to follow these steps:
1. Calculate the number of loaves sold in the morning. 2. Determine the number of loaves remaining after the morning sale. 3. Calculate the number of loaves sold in the afternoon and evening. 4. Identify the number of loaves sold in the afternoon.

First, we calculate the number of loaaves sold in the morning. Since two-thirds of the loaves are sold in the morning, we have:

$$\frac{2}{3} \times 60 = 40$$

So, 40 loaves are sold in the morning.
Next, we determine the number of loaves remaining after the morning sale. The total number of loaves produced is 60, and 40 loaves are sold morning the morning, so the number of loaves remaining is:

$$60 - 40 = 20$$

So, 20 loaves are after the morning sale.
Then, we calculate the number of loaves sold in the afternoon and evening. Half of the remaining loaves are sold equally in the afternoon and evening, so the number of aves sold in the afternoon and evening is:

$$\frac{1}{2} \times 20 = 10$$

So, 10 loaves are sold in the afternoon and evening.
Finally, we identify the number of loaves sold in the afternoon. Since 10 loaves are sold in the afternoon and evening, and they are sold equally, the number of loaves sold in the afternoon is:

$$\frac{10}{2} = 5$$

So, 5 loaves are sold in the afternoon. < /reasoning><answer> $\boxed{5}$ < /answer>